

Lab 3: Sequential Logic and Counters

Requirements Not Met

N/A

Problems Encountered

Quartus was difficult to manage when implementing the flip flops.

Applications

This lab demonstrates the practical applications of digital counters, flip-flops, and combinational logic in digital systems. The bidirectional counter can be used in systems that need to track events or positions in both directions, such as digital clocks, frequency counters, motor position control, and up/down counting systems. The seven-segment decoder also demonstrates how binary state information can be converted into a human-readable hexadecimal display.

Pre-Lab Questions

1. When implementing an SR latch, what happens when both $S=1$ and $R=1$?

When $S=1$ and $R=1$, the SR latch enters an invalid state because both outputs are forced to 0. This condition should normally be avoided.

2. What is the purpose of the Clk signal?

The Clk signal controls when a flip-flop is allowed to change its output. It keeps changes in the circuit synchronized.

3. What is the difference between a SR flip-flop and a SR latch?

An SR latch is level-sensitive and can change while it is enabled. An SR flip-flop is edge-triggered and changes only on a clock edge.

4. Put a NOT gate in front of the S input of the SR flip-flop, then connect the input of that gate to the R input of the flip-flop so that S and R are always at opposite logic levels. What is the relationship between this single input and the output Q?

If the single input is X, then $S = \text{NOT } X$ and $R = X$. Therefore, the output is $Q = \text{NOT } X$.

5. Now do the same thing, except put the NOT gate in front of R instead. What is the relationship between the single input and Q?

If the single input is X, then $S = X$ and $R = \text{NOT } X$. Therefore, the output is $Q = X$.

6. What are don't-care entries in an NSTT and why are they useful when simplifying logic?

Don't-care entries are input combinations where the output does not matter, usually represented by X. They are useful because they can be treated as either 0 or 1 to simplify the logic.

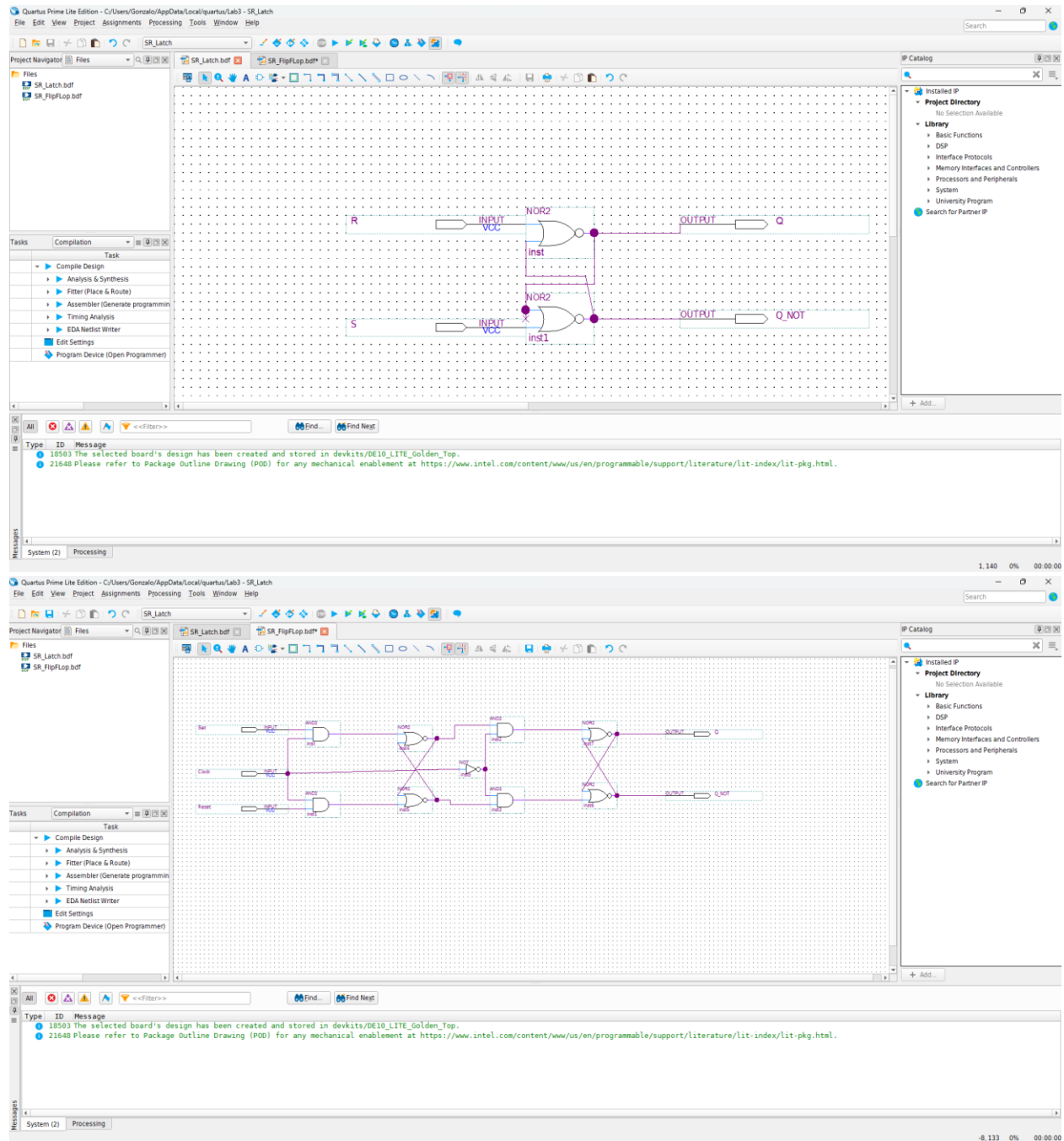
7. Why do we need flip-flops to build a counter? Why can't we build one with only combinational logic?

Flip-flops are needed because a counter must remember its current state. Combinational logic does not have memory, so it cannot store the previous count.

8. Describe D, SR, JK, and T flip-flops: what inputs does each use, and how does Q change on a clock edge for each?

A D flip-flop uses D and sets Q equal to D, while an SR flip-flop uses S and R to set or reset Q.

A JK flip-flop uses J and K to set, reset, hold, or toggle Q, while a T flip-flop uses T to hold Q when $T=0$ or toggle Q when $T=1$.



Part 2:

Q1	Q0	GO(H)	Q1+	Q0+
0	0	0	X	X
0	0	1	X	X

0	1	0	0	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	1
1	1	0	1	1
1	1	1	1	0

DO	GO	
Q/Q	0	1
00	X	X
01	1	1
10	1	0
11	0	1

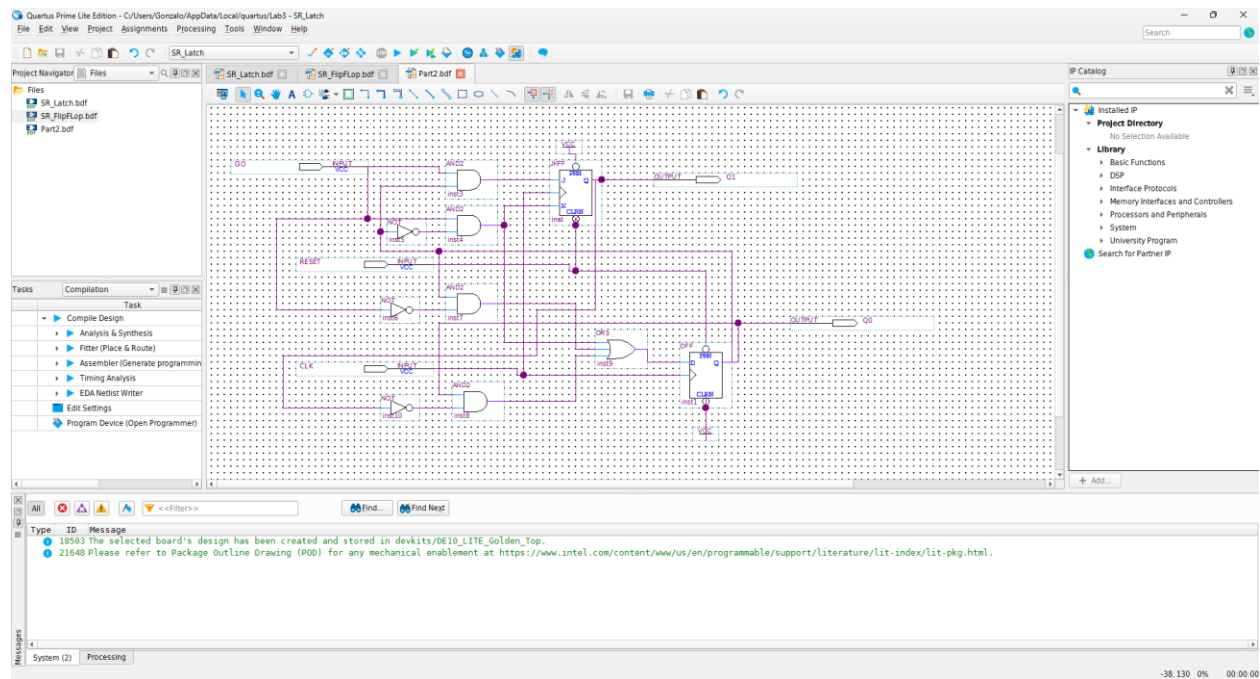
J1	GO	
Q/Q	0	1
00	0	X
01	0	1
11	X	X
10	X	X

$$P_0 = \bar{Q}_1 + \bar{Q}_0 Q_0 + G_0 \bar{Q}_0$$

$$\bar{J}_1 = G_0$$

K1	GO	0	1
Q/Q	0	X	X
00	X	X	
01	X	X	
11	0	0	
10	0	1	

$$K_1 = G_0 Q_0$$



Part 3

Q2	Q1	Q0	F	Q2+	Q1+	Q0+	Y3	Y2	Y1	Y0
1	0	0	1	1	0	1	0	1	0	0
1	0	0	0	1	1	0	0	1	0	0
1	0	1	1	0	1	0	0	1	0	1
1	0	1	0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	1	0	1	0
0	1	0	0	1	0	1	1	0	1	0
0	1	1	1	1	1	0	1	0	1	1
0	1	1	0	0	1	0	1	0	1	1
1	1	0	1	1	0	0	0	0	1	0
1	1	0	0	0	1	1	0	0	1	0
0	0	0	1	X	X	X	X	X	X	X
0	0	0	0	X	X	X	X	X	X	X
0	0	1	1	X	X	X	X	X	X	X
0	0	1	0	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X	X

Q2Q1Q0	Hex output	Y3Y2Y1Y0
100	4	0100
101	5	0101
010	A	1010

011	B	1011
110	2	0010
000	unused	XXXX
001	unused	XXXX
111	unused	XXXX

$J_2 = Q_0 F$				$K_2 = Q_0 F + Q_0 F$			
Q_2	Q_1	Q_0	F	Q_2	Q_1	Q_0	F
00	01	11	10	00	01	11	10
00	X	X	X	00	X	X	X
01	X	1	0	01	X	X	0
11	X	X	X	11	0	X	0
10	X	X	X	10	1	1	0

T_1		$Q_0 F$			
$Q_2 Q_1$	Q_0	Q_0	Q_1	11	10
00	0	x	x	x	10
01	1	0	0	0	0
11	0	0	x	1	x
10	1	0	1	1	x

$$T_1 = Q_2 Q_0 + Q_2 Q_1 F + \overline{Q_2} \overline{Q_1} \overline{F} + Q_2 Q_1 F$$

D_0		$Q_0 F$			
$Q_2 Q_1$	Q_0	Q_0	Q_1	11	10
00	0	x	x	x	x
01	1	1	0	0	0
11	1	0	x	x	x
10	0	1	0	0	0

$$D_0 = \overline{Q_2} \overline{Q_0} + Q_1 \overline{Q_0} F + \overline{Q_1} Q_0 F$$

$$Y_3 = Q_1 \quad Y_2 = Q_2 \quad Y_1 = Q_1$$

$$Y_0 = Q_2 \overline{Q_1} Q_0 + \overline{Q_2} Q_1 Q_0$$